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SOLID-STATE IMAGING DEVICE

Abstract

Provided is a solid-state imaging device that enables reduction of color mixing and improvement of light absorbency in a photoelectric conversion unit. A pixel includes a first periodic structure, the first periodic structure includes a plurality of first layers and a plurality of second layers having a lower refractive index than that of the first layers, and in at least one of a plurality of pixels, the first layers and the second layers of the first periodic structure are arranged such that, in a pixel central portion in the horizontal direction, the volume proportions of the first layers in the horizontal direction are greater than the volume proportions of the second layers, and the volume proportions of the first layers decrease from the pixel central portion toward a pixel peripheral portion in the horizontal direction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority under 35 U.S.C. § 119 to Japanese Patent Application No. 2024-032693, filed on Mar. 5, 2024, in the Japan Patent Office, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

[0002] The inventive concept relates to a solid-state imaging device.

[0003] Solid-state imaging devices are installed in various mobile terminals such as digital cameras or mobile phones.

[0004] Referring to Japanese Patent Publication No. 2022-108744, a solid-state imaging device includes a photoelectric conversion unit configured to generate and accumulate electric charges according to the amount of incident light, an on-chip lens installed on one side of the photoelectric conversion unit where light is incident, and a wiring layer installed on the other side of the photoelectric conversion unit.

[0005] In a solid-state imaging device, an annular portion (zone plate) having a concave-convex shape is provided on one side of a photoelectric conversion unit, wherein the annular portion includes a first annular portion including a semiconductor region (a part of a P-type semiconductor region), and a second annular portion including a fixed charge film and an insulating film arranged to surround the first annular portion. The annular portion having a concave-convex shape collects light incident through the on-chip lens, onto the photoelectric conversion unit through a lens effect. Thus, the solid-state imaging device includes the annular portion that has a concave-convex shape and exhibits a lens effect, and thus may reduce the occurrence of color mixing between pixels.

SUMMARY

[0006] One of indicators for evaluating the performance of a solid-state imaging device is the light absorbency in a photoelectric conversion unit. A solid-state imaging device exhibits high quantum efficiency (photoelectric conversion efficiency) as a photoelectric conversion unit has higher light absorbency.

[0007] As described above, the solid-state imaging device may reduce color mixing by installing an additional structure, such as an annular portion having a concave-convex shape, in the photoelectric conversion unit. However, the solid-state imaging device of Japanese patent publication No. 2022-108744 does not specifically mention that it seeks to improve the quantum efficiency of the photoelectric conversion unit by focusing research on the structure of the photoelectric conversion unit. Therefore, it may be seen that the solid-state imaging device has room for further improvement in the quantum efficiency of the photoelectric conversion unit.

[0008] The inventive concept has been made to solve the above issue, and is to provide a solid-state imaging device that enables reduction of color mixing and improvement of light absorbency in a photoelectric conversion unit.

[0009] According to an aspect of the inventive concept, there is provided a solid-state imaging device including a pixel array including a plurality of pixels, wherein each of the plurality of pixels includes a photoelectric conversion unit configured to convert light into an electric charge, an on-chip lens installed on one side of the photoelectric conversion unit, wiring layers installed on another side of the photoelectric conversion unit, and a first periodic structure that is installed on the one side of the photoelectric conversion unit and has periodicity in a horizontal direction that is perpendicular to a stacking direction of the photoelectric conversion unit, the first periodic structure includes a plurality of first layers and a plurality of second layers that have refractive indices lower than refractive indices of the plurality of first layers, and in at least one of the plurality of pixels, the first layers and the second layers of the first periodic structure are arranged such that volume

proportions of the first layers in a pixel central portion in the horizontal direction are greater than volume proportions of the second layers in the pixel central portion, and the volume proportions of the first layers decrease from the pixel central portion toward a pixel peripheral portion in the horizontal direction.

[0010] In an embodiment, widths of the second layers in the horizontal direction may be different from each other, and the first periodic structure may be configured such that the volume proportions of the first layers decrease from the pixel central portion toward the pixel peripheral portion.

[0011] In an embodiment, depths of the second layers in the stacking direction may be different from each other, and the first periodic structure may be configured such that the volume proportions of the first layers decrease from the pixel central portion toward the pixel peripheral portion.

[0012] In an embodiment, a period of the first periodic structure may be set to a certain value based on a wavelength and an angle of incidence of light received by the photoelectric conversion unit, and may include a length that is less than the wavelength of the received light, and enables generation of diffracted light in the photoelectric conversion unit.

[0013] In an embodiment, depths of the first layers and depths of the second layers in the stacking direction may be equal to a wavelength of light received by the photoelectric conversion unit, and a thickness of the photoelectric conversion unit may be greater than the depths of the first layers and the depths of the second layers.

[0014] In an embodiment, the photoelectric conversion unit may be covered at least in part by a dielectric layer having a refractive index lower than the refractive indices of the first layers, and a period of the first periodic structure may include a length that enables diffracted light, which is generated in the photoelectric conversion unit, to be totally reflected at a boundary between the photoelectric conversion unit and the dielectric layer.

[0015] In an embodiment, the wiring layer, which is arranged at a position closest to the photoelectric conversion unit in the stacking direction, may have a reflective structure that reflects light transmitted through the photoelectric conversion unit.

[0016] In an embodiment, the pixel including the first periodic structure may further include a second periodic structure that is installed on the one side of the photoelectric conversion unit and has periodicity in the horizontal direction, and the second periodic structure may include a plurality of third layers having refractive indices equal to refractive indices of the first layers, and a plurality of fourth layers having refractive indices equal to refractive indices of the second layers.

[0017] In an embodiment, the second periodic structure may have periodicity that is identical to the periodicity of the first periodic structure.

[0018] In an embodiment, distribution of the volume proportions of the first layers in the first periodic structure may be identical to distribution of volume proportions of the third layers in the second periodic structure, and distribution of the volume proportions of the second layers in the first periodic structure may be identical to distribution of volume proportions of the fourth layers in the second periodic structure.

[0019] According to an aspect of the inventive concept, there is provided a solid-state imaging device comprising a pixel array comprising a plurality of pixels, wherein each of the plurality of pixels comprises: a photoelectric conversion unit configured to convert light into an electric charge; an on-chip lens installed on one side of the photoelectric conversion unit; a wiring layer installed on another side of the photoelectric conversion unit; and a first periodic structure that is installed on the one side of the photoelectric conversion unit and has periodicity in a horizontal direction that is perpendicular to a stacking direction of the photoelectric conversion unit, wherein the first periodic structure comprises a plurality of first layers and a plurality of second layers that have refractive indices lower than refractive indices of the plurality of first layers, wherein in at least one of the plurality of pixels, the first layers and the second layers of the first periodic structure are arranged

such that volume proportions of the first layers in a pixel central portion are greater than volume proportions of the second layers in the pixel central portion, and the volume proportions of the first layers decrease from the pixel central portion toward a pixel peripheral portion in the horizontal direction, wherein the pixel comprising the first periodic structure further comprises a second periodic structure that is installed on the one side of the photoelectric conversion unit and has periodicity in the horizontal direction, and wherein the second periodic structure comprises a plurality of third layers having refractive indices equal to the refractive indices of the first layers, and a plurality of fourth layers having refractive indices equal to the refractive indices of the second layers.

[0020] According to an aspect of the inventive concept, there is provided a solid-state imaging device comprising a pixel array comprising a plurality of pixels, wherein each of the plurality of pixels comprises: a photoelectric conversion unit configured to convert light into an electric charge; an on-chip lens installed on one side of the photoelectric conversion unit; a wiring layer installed on another side of the photoelectric conversion unit; and a first periodic structure that is installed on the one side of the photoelectric conversion unit and has periodicity in a horizontal direction that is perpendicular to a stacking direction of the photoelectric conversion unit, wherein the first periodic structure comprises a plurality of first layers and a plurality of second layers that have refractive indices lower than refractive indices of the plurality of first layers, and wherein in at least one of the plurality of pixels, depths of the second layers of the first periodic structure are equal to each other, and widths of the second layers increase from a pixel central portion toward a pixel peripheral portion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] Embodiments will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings in which:

[0022] FIG. 1 is a block diagram illustrating a solid-state imaging device according to an embodiment;

[0023] FIG. 2 is a plan view illustrating a first periodic structure of a pixel according to a first embodiment;

[0024] FIG. 3 is a cross-sectional view of pixels according to the first embodiment, taken along a stacking direction;

[0025] FIG. 4A is a cross-sectional view illustrating a first periodic structure of a pixel that is arranged at a pixel array central portion, according to the first embodiment;

[0026] FIG. 4B is a cross-sectional view illustrating a first periodic structure of a pixel that is arranged at a pixel array peripheral portion, according to the first embodiment;

[0027] FIG. 5 is a diagram for describing a functional effect of a first periodic structure of a pixel, according to the first embodiment;

[0028] FIG. 6 is a diagram for describing a functional effect of a first periodic structure of a pixel, according to the first embodiment;

[0029] FIG. 7 is a cross-sectional view for describing a functional effect of a first periodic structure of a pixel, according to the first embodiment;

[0030] FIG. 8 is a cross-sectional view illustrating a first periodic structure according to modified example 1;

[0031] FIG. 9 is a cross-sectional view illustrating a first periodic structure according to modified example 2;

[0032] FIG. 10 is a cross-sectional view illustrating a first periodic structure according to modified example 3;

[0033] FIG. **11** is a plan view illustrating a first periodic structure according to modified example [0034] 4;

[0035] FIG. **12** is a plan view illustrating a first periodic structure according to modified example 5;

[0036] FIG. **13** is a plan view illustrating a first periodic structure according to modified example 6;

[0037] FIG. **14** is a plan view illustrating a first periodic structure according to modified example 7;

[0038] FIG. **15** is a plan view illustrating a first periodic structure according to modified example 8;

[0039] FIG. **16** is a cross-sectional view of the pixels according to a second embodiment, taken along a stacking direction;

[0040] FIG. **17** is a cross-sectional view for describing a functional effect of a first periodic structure of a pixel, according to the second embodiment;

[0041] FIG. **18** is a cross-sectional view of pixels according to a third embodiment, taken along a stacking direction;

[0042] FIG. **19** is a diagram for describing a functional effect of a first periodic structure of a pixel, according to the third embodiment;

[0043] FIG. **20** is a cross-sectional view of pixels according to a fourth embodiment, taken along a stacking direction;

[0044] FIG. **21** is a diagram for describing a functional effect of a first periodic structure of a pixel, according to the fourth embodiment;

[0045] FIG. **22** is a cross-sectional view of pixels according to a fifth embodiment, taken along a stacking direction; and

[0046] FIG. **23** is a diagram for describing a functional effect of a first periodic structure of a pixel, according to the fifth embodiment.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0047] In the drawings, the size and relative sizes of layers and regions may be exaggerated for clarity. Like reference characters refer to like elements throughout. Though the different figures show variations of exemplary embodiments, these figures are not necessarily intended to be mutually exclusive from each other. Rather, as will be seen from the context of the detailed description below, certain features depicted and described in different figures can be combined with other features from other figures to result in various embodiments, when taking the figures and their description as a whole into consideration.

First Embodiment

[0048] Hereinafter, a first embodiment will be described with reference to FIGS. **1** to **7**. In addition, a dimensional ratio in the drawings may be exaggerated for convenience of description, and may differ from the actual ratio.

[0049] FIG. **1** is a block diagram schematically illustrating an overall configuration of a solid-state imaging device **1** according to the present embodiment. FIG. **2** is a plan view illustrating a single pixel **10** provided in the solid-state imaging device **1**. FIG. **3** is a cross-sectional view of the pixels **10** illustrated in FIG. **2**, taken along a stacking direction. FIG. **4A** is a cross-sectional view schematically illustrating a portion of a first periodic structure **140** arranged in a central portion **21** of a pixel array. FIG. **4B** is a cross-sectional view schematically illustrating a portion of the first periodic structure **140** arranged in a peripheral portion **23** of a pixel array. FIGS. **5** to **7** are diagrams for describing functional effects of the first periodic structure **140**.

[0050] The solid-state imaging device **1** according to the present embodiment is configured as a solid-state imaging device in the form of a complementary metal-oxide-semiconductor (CMOS) device.

[0051] As illustrated in FIGS. **1** and **2**, the solid-state imaging device **1** includes a pixel array **20**

consisting of a plurality of pixels **10** configured to output pixel signals, a control circuit **30** configured to generate an operating signal for operating each unit pixel of the solid-state imaging device **1**, a vertical driver circuit **40** configured to vertically scan each pixel **10** and control the output of a pixel signal according to the amount of light received by each pixel **10**, a horizontal driver circuit **50** configured to output a horizontal scanning pulse, a column signal processing circuit **60** configured to process a pixel signal output from each pixel **10** to generate an image signal, a vertical signal line **70** configured to transmit a pixel signal generated from each pixel **10** to the column signal processing circuit **60**, a horizontal signal line **80** configured to output an image signal from the column signal processing circuit **60**, and an output circuit **90** configured to process an image signal received through the horizontal signal line **80** and output the processed signal.

[0052] For components of the solid-state imaging device **1** other than the pixels **10**, any known configuration in the technical field of solid-state imaging devices may be arbitrarily and selectively employed. Thus, in the present specification, descriptions of components other than the pixels **10** will be appropriately omitted.

[0053] As illustrated in FIG. **1**, the plurality of pixels **10** are regularly arranged on a substrate **25**. The pixel array **20** consists of the plurality of pixels **10** arranged in a two-dimensional array in the plan view of FIG. **1**. The substrate **25** on which the pixels **10** are arranged is a semiconductor substrate, such as a silicon (Si) substrate. As used herein, the term “pixel” or “unit pixel” refers to a sensor element of an image sensor, and may refer to a smallest addressable light-sensing element of the image sensor.

[0054] As illustrated in FIGS. **2** and **3**, the solid-state imaging device **1** includes a photoelectric conversion unit **110** installed in each pixel **10**. In addition, FIG. **3** illustrates an example of two pixels **10** adjacent to each other in a horizontal direction.

[0055] In the present specification, a direction in which an on-chip lens **120** and the photoelectric conversion unit **110** are stacked (a direction indicated by the arrow **Z1** and **Z2** in FIG. **3**) is referred to as the “stacking direction.” In addition, a direction orthogonal to the stacking direction (a direction parallel to each arrow **X1**, **X2**, **Y1**, and **Y2** in FIG. **2**) is referred to as a “horizontal direction.”

[0056] As illustrated in FIGS. **2** and **3**, the pixel **10** includes the photoelectric conversion unit **110** configured to convert light into an electric charge, the on-chip lens **120** installed on one side of the photoelectric conversion unit **110** (the side indicated by the arrow **Z1**, hereinafter, also simply referred to as ‘**Z1** side’), wiring layers **131** and **132** installed on the other side of the photoelectric conversion unit **110** (the side indicated by the arrow **Z2**, hereinafter also simply referred to as ‘**Z2** side’), and a first periodic structure **140** that is installed on the **Z1** side of the photoelectric conversion unit **110** and has periodicity in the horizontal direction of the photoelectric conversion unit **110**. As discussed further below, the first periodic structure **140** includes a plurality of first layers **141** and a plurality of second layers **142**.

[0057] The solid-state imaging device **1** is configured as a so-called back-illuminated type. Thus, the on-chip lens **120** is arranged on the back side of the pixel **10** (the same side as the **Z1** side). Light **F** incident on the pixel **10** enters the photoelectric conversion unit **110** from the **Z1** side through the on-chip lens **120**.

[0058] In the drawings, for convenience of description, respective components of the light **F** incident on the on-chip lens **120** are assigned different symbols **F1**, **F2**, and **F3** according to their positions of incidence on the photoelectric conversion unit **110**. In the present specification, the entire incident light is simply referred to as ‘incident light **F**’.

[0059] The photoelectric conversion unit **110** includes a p-type semiconductor region and an n-type semiconductor region. In the photoelectric conversion unit **110**, a photodiode is implemented by a p-n junction between the p-type semiconductor region and the n-type semiconductor region, and the photodiode converts light into an electric charge. The photoelectric conversion unit **110** receives light incident through the on-chip lens **120**, generates electric charges in correspondence to

the amount of light received, and accumulates the generated electric charges in the n-type semiconductor region.

[0060] As illustrated in FIGS. 3, 4A, and 4B, the first periodic structure **140** is installed within a certain range in the stacking direction of the photoelectric conversion unit **110**. In the present embodiment, as illustrated in FIGS. 4A and 4B, the first periodic structure **140** corresponds to a range in which second layers **142** extend in the stacking direction of the photoelectric conversion unit **110** (a range between an end of the second layer **142** on the Z1 side and another end of the second layer **142** on the Z2 side), and includes a region (part) in which layers **141** and **142** are alternately arranged in the horizontal direction. In some example embodiments, the first periodic structure **140** may have a thickness (or height) in the stacking direction that is equal to the thickness (or height) of the second layers **142** in the stacking direction.

[0061] As illustrated in FIG. 3, the photoelectric conversion units **110**, which are adjacent to each other in the horizontal direction, are separated from each other by an insulating film **170**. In example embodiments, a lower surface of the insulating film **170** may be at the same level as lower surfaces of the adjacent photoelectric conversion unit **110**, and an upper surface of the insulating film **170** may be at a higher level than upper surfaces of the adjacent photoelectric conversion unit **110**. As the pixel **10** includes the insulating film **170**, when a signal charge exceeding a saturation charge amount occurs, the excess signal charge may be prevented from flowing out from the photoelectric conversion unit **110** to another photoelectric conversion unit **110** adjacent thereto in the horizontal direction. As a constituent material of the insulating film **170**, for example, a silicon oxide film, a silicon nitride film, a silicon oxynitride film, a resin film, and the like may be used. In addition, as the insulating film **170**, a film without positive fixed charges or a film with a small amount of positive fixed charges may be used.

[0062] A fixed charge film may be installed between the photoelectric conversion units **110** that are adjacent to each other in the horizontal direction. As the fixed charge film is installed in the pixel **10**, the pixel **10** may reduce noise by reducing the occurrence of dark current. As a constituent material of the fixed charge film, for example, an oxide film or a nitride film including at least one metal element from among hafnium (Hf), aluminum (Al), zirconium (Zr), tantalum (Ta), and titanium (Ti) is used. Methods of forming the fixed charge film include, for example, chemical vapor deposition (CVD), sputtering, atomic layer deposition (ALD), and the like.

[0063] The on-chip lens **120** collects the incident light F entering from the Z1 side. The light collected by the on-chip lens **120** enters the photoelectric conversion unit **110**.

[0064] As illustrated in FIG. 2, a central portion **121** of the on-chip lens **120** in the horizontal direction is arranged to overlap with a pixel central portion **11** of the pixel **10**.

[0065] As illustrated in FIG. 3, the solid-state imaging device **1** includes a plurality of wiring layers **131** and **132** arranged with intervals therebetween in the stacking direction. Hereinafter, the wiring layer **131** arranged closest to the photoelectric conversion unit **110** is referred to as a 'first wiring layer', and the wiring layer **132** arranged further away from the photoelectric conversion unit **110** than the first wiring layer **131** is referred to as a 'second wiring layer'. In addition, there are no particular limitations on the number of wiring layers installed in one solid-state imaging device **1**, their arrangement in the horizontal direction, and their cross-sectional shape in the stacking direction, and they may be arbitrarily modified.

[0066] The first wiring layer **131** and the second wiring layer **132** extract, as pixel signals, signal charges generated and accumulated by the photoelectric conversion unit **110**. The first wiring layer **131** and the second wiring layer **132** output the extracted pixel signals through the vertical signal line **70**.

[0067] As illustrated in FIGS. 2 and 3, the first periodic structure **140** includes a plurality of first layers **141**, and a plurality of second layers **142** having a lower refractive index than the first layers **141**.

[0068] A 'difference in refractive index' between the first layer **141** and the second layer **142** may

be defined, for example, by the physical properties (e.g., dielectric constant) of the constituent materials of the respective layers **141** and **142**. In addition, there are no particular limitations on the ‘difference in refractive index’ as long as a lens effect to be described below is exhibited, and diffracted light Fr to be described below is generated inside the photoelectric conversion unit **110**. [0069] As constituent materials of the respective layers **141** and **142** for realizing a desired difference in refractive index, for example, a combination of the following materials may be selected.

[0070] As a constituent material of the first layer **141**, for example, any one of silicon (Si), germanium (Ge), and indium gallium arsenide (InGaAs) may be used. For example, the first layer **141** may be formed of or include any one of silicon (Si), germanium (Ge), and indium gallium arsenide (InGaAs).

[0071] In a case in which the first layer **141** is formed of any one of the above materials, for example, any one of silicon dioxide (SiO₂), silicon nitride (SiN), aluminum oxide (AlO), tantalum oxide (TaO), titanium nitride (TiN), and titanium oxide (TiO) may be used as a constituent material of the second layer **142**. For example, the second layer **142** may be formed of or include any one of silicon dioxide (SiO₂), silicon nitride (SiN), aluminum oxide (AlO), tantalum oxide (TaO), titanium nitride (TiN), and titanium oxide (TiO).

[0072] The first periodic structure **140** is configured to have periodicity in the horizontal direction as described above. The expression ‘having periodicity in the horizontal direction’ means that the arrangement interval of the second layers **142** in the horizontal direction (hereinafter, referred to as ‘period p’) is constant over the entire range in the horizontal direction or an arbitrary partial range in the horizontal direction, as illustrated in FIGS. **2**, **4A**, and **4B**.

[0073] As illustrated in FIGS. **4A** and **4B**, in the pixel **10** of the present embodiment, the period p is constant over the entire range in a direction parallel to the horizontal arrow X1 and X2. However, referring to FIG. **1**, the period p of the first periodic structure **140** may also be configured differently in the central portion **21** of the pixel array (an arbitrary region including a center position in a plane direction of the pixel array **20**), and in the peripheral portion **23** of the pixel array (an arbitrary region outside the central portion **21** of the pixel array in plan view). For example, a period p1 of the first periodic structure **140** arranged in the central portion **21** (see FIG. **1**) of the pixel array illustrated in FIG. **4A** may be less than a period p2 of the first periodic structure **140** arranged in the peripheral portion **23** (see FIG. **1**) of the pixel array illustrated in FIG. **4B**. As the length of the period p varies in the central portion **21** (see FIG. **1**) of the pixel array and the peripheral portion **23** (see FIG. **1**) of the pixel array, angles at which light incident on the first periodic structure **140** at the central portion **21** (see FIG. **1**) of the pixel array and the peripheral portion **23** (see FIG. **1**) is diffracted on the surface of the first layer **141** are equivalent to each other, the diffracted light is totally reflected at a boundary between the photoelectric conversion unit **110** and the insulating film **170** that are adjacent to each other in the horizontal direction, and the light is confined within the photoelectric conversion unit **110**, enabling the effect of achieving high quantum efficiency in both the central portion **21** (see FIG. **1**) of the pixel array and the peripheral portion **23** (see FIG. **1**). In addition, in the following description, operation effects according to the present embodiment will be described based on the first periodic structure **140** arranged in the central portion **21** (see FIG. **1**) of the pixel array illustrated in FIG. **4A**.

[0074] FIG. **4A** illustrates an example of four second layers **142a**, **142b**, **142c**, and **142d** arranged with the period p therebetween, from the pixel central portion **11** of the pixel **10** of FIG. **1** toward a pixel peripheral portion **13** of the pixel **10** in the horizontal direction. For example, FIG. **4A** illustrates four second layers **142a**, **142b**, **142c**, and **142d** arranged in order in a direction parallel to the horizontal arrow X1 and four second layers **142a**, **142b**, **142c**, and **142d** arranged in order in a direction parallel to the horizontal arrow X2. However, there are no particular limitations on the number of second layers **142** provided in the pixel **10**.

[0075] In the plan view of FIG. **2**, the first layers **141** are arranged in a quadrangular pattern with a

certain area.

[0076] In the plan view of FIG. 2, the second layers **142** are arranged in a line pattern extending to space out the first layers **141** that are adjacent to each other in the horizontal direction.

[0077] In the cross-sectional views of FIGS. 3 and 4A, a concave-convex structure formed by the first layers **141** and the second layers **142** is installed on a surface of the photoelectric conversion unit **110** on the Z1 side. The concave-convex structure is formed by the second layers **142** having a rectangular cross-sectional shape and extending from the Z1 side of the photoelectric conversion unit **110** to penetrate into the first layers **141**.

[0078] FIG. 4A illustrates a direction parallel to the arrows X1 and X2, as a horizontal direction that serves as a reference for the period p of the first periodic structure **140**. However, the horizontal direction that serves as a reference for the periodicity of the first periodic structure **140** is not limited to a direction parallel to the arrows X1 and X2. For example, the first periodic structure **140** may be configured to have a period p in at least one of the direction of an arrow A1 passing through the pixel central portion **11** and orthogonal to the arrow X1 and X2 in the plan view of FIG. 2, the direction of an arrow A2 passing through the pixel central portion **11** and orthogonal to the arrow Y1 and Y2, and the directions of arrows A3 and A4 passing through the pixel central portion **11** and intersecting the arrows A1 and A2 at 45° in the plan view.

[0079] As illustrated in FIG. 2, the first layer **141** is arranged in a central portion **145** of the first periodic structure **140** in the horizontal direction. The central portion **145** of the first periodic structure **140** in the horizontal direction is arranged to overlap with the pixel central portion **11**.

[0080] In the solid-state imaging device **1**, the first periodic structure **140** in at least one of the plurality of pixels **10** has the following configuration.

[0081] As illustrated in FIGS. 2, 3, and 4A, in the first periodic structure **140**, the first layers **141** and the second layers **142** are arranged such that, in the pixel central portion **11** in the horizontal direction, the volume proportions of the first layer **141** are greater than the volume proportions of the second layer **142**, and the volume proportions of the first layers **141** decrease from the pixel central portion **11** toward the pixel peripheral portion **13** in the horizontal direction. In addition, the first layers **141** and the second layers **142** are arranged such that the volume proportions of the second layers **142** increase from the pixel central portion **11** toward the pixel peripheral portion **13** in the horizontal direction. For example, when viewed in cross-section, the width of the first layer **141** becomes progressively narrower as the distance increases from the pixel central portion **11**, and the width of the second layer becomes progressively wider as the distance increases from the pixel central portion **11**.

[0082] Regarding the term ‘volume proportion’, for example, the volume proportion of the first layer **141** refers to the proportion of the volume of the first layer **141** with respect to the total volume of the photoelectric conversion unit **110**, and the volume proportion of the second layer **142** refers to the proportion of the volume of the second layer **142** with respect to the total volume of the photoelectric conversion unit **110**.

[0083] The functional effects of the first periodic structure **140** (collection of incident light through a lens effect) will be described with reference to FIG. 5.

[0084] (a) of FIG. 5 is a graph showing a relationship between the magnitude of an average refractive index for lights F1, F2, and F3 incident on the photoelectric conversion unit **110**, and horizontal positions in the photoelectric conversion unit **110** (the first periodic structure **140**). (b) of FIG. 5 is a color bar showing a magnitude relationship between the refractive indices for the incident lights F1, F2, and F3 indicated to correspond to the horizontal positions in the graph. (c) of FIG. 5 schematically illustrates a principle in which each of the incident lights F1, F2, and F3 is collected by the first periodic structure **140**. A straight line H1 illustrated in (a) to (c) of FIG. 5 indicates a position in the first periodic structure **140** where the refractive index is the greatest (the same position as the position of the pixel central portion **11** and the position of the central portion **145** of the first periodic structure **140** in FIG. 2).

[0085] In the first periodic structure **140**, the volume proportion of the first layer **141** in the pixel central portion **11** is greater than the volume proportion of the second layer **142**. In addition, the refractive index of the first layer **141** is greater than the refractive index of the second layer **142**. Thus, as illustrated in (a) of FIG. 5, the refractive index of the first periodic structure **140** in the horizontal direction is the greatest in the pixel central portion **11**.

[0086] In the first periodic structure **140**, the first layers **141** and the second layers **142** are arranged such that the volume proportions of the first layers **141** decrease from the pixel central portion **11** toward the pixel peripheral portion **13**. Thus, as illustrated in (a) of FIG. 5, the refractive index of the first periodic structure **140** in the horizontal direction gradually decreases from the pixel central portion **11** toward the pixel peripheral portion **13**.

[0087] The color bar of (b) of FIG. 5 indicates a magnitude relationship in the refractive index distribution in (a) of FIG. 5. In the color bar, a position with a higher refractive index is represented by a lighter color, and a position with a lower refractive index is represented by a darker color. As illustrated in (b) of FIG. 5, a position corresponding to the vicinity of the pixel central portion **11** where the volume proportion of the first layer **141** is greater than the volume proportion of the second layer **142** has a high refractive index. Thus, in the color bar, the position corresponding to the vicinity of the pixel central portion **11** is represented by a color close to white. A position corresponding to the vicinity of the pixel peripheral portion **13** where the volume proportion of the first layer **141** is less than that of the pixel central portion **11** has a low refractive index. Thus, in the color bar, the position corresponding to the vicinity of the pixel peripheral portion **13** is represented by a color close to black.

[0088] As illustrated in (c) of FIG. 5, the incident light **F1** (actually, some components of the incident light), which has passed through the vicinity of the pixel central portion **11** (the central portion **145** of the first periodic structure **140**) and then entered the photoelectric conversion unit **110**, passes through a portion of the first periodic structure **140** where the refractive index is high. Thus, the velocity at which the incident light **F1** propagates into the first periodic structure **140** decreases.

[0089] As illustrated in (c) of FIG. 5, the incident lights **F2** and **F3** (actually, some components of incident light), which have passed through the pixel peripheral portion **13** and then entered the photoelectric conversion unit **110**, passes through a portion of the first periodic structure **140** where the refractive index is less than that in the pixel central portion **11**. Thus, when the incident lights **F2** and **F3** enter the photoelectric conversion unit **110** from the first periodic structure **140**, the velocity at which the incident lights **F2** and **F3** propagate into the first periodic structure **140** increases.

[0090] As described above, the incident light **F1** differs from the incident lights **F2** and **F3** in the velocity of propagation, and thus, the incident lights **F2** and **F3** propagate while refracting inside the first periodic structure **140**, like a wavelength component **WA** schematically illustrated in (c) of FIG. 5. As such, the first periodic structure **140** exhibits a lens effect that collects incident light passing through the first periodic structure **140**, onto the first wiring layer **131** arranged on the **Z2** side.

[0091] In the pixel **10**, the first periodic structure **140** may exhibit a lens effect to efficiently allow incident light that has passed through the photoelectric conversion unit **110** to reach the first wiring layer **131** arranged on the **Z2** side of the photoelectric conversion unit **110**. Thus, the pixel **10** may prevent incident light, which has passed through the photoelectric conversion unit **110**, from passing through spaces between the first wiring layers **131** that are adjacent to each other in the horizontal direction (areas where no first wiring layers **131** are arranged in the cross-sectional view of FIG. 3), thereby suppressing the occurrence of color mixing.

[0092] The functional effects of the first periodic structure **140** (improvement of quantum efficiency by diffracted light) will be described with reference to FIG. 6. FIG. 6 is a schematic diagram for describing a principle of generation of diffracted light. Thus, the arrangement or cross-

sectional shape of the first layers **141** and the second layers **142** illustrated in FIG. **6** do not have a strict correspondence with the embodiment illustrated in FIG. **3**.

[0093] The first layers **141** and the second layers **142** are regularly arranged such that the first periodic structure **140** has a period p in the horizontal direction. As illustrated in FIG. **6**, the incident light F enters the first periodic structure **140** from the $Z1$ side of the photoelectric conversion unit **110**. The first periodic structure **140** generates diffracted light Fr . The first periodic structure **140** has a constant period p in the horizontal direction. Thus, the first periodic structure **140** may generate the diffracted light Fr in each section in the horizontal direction according to the period p .

[0094] As illustrated in FIGS. **3** and **6**, the diffracted light Fr generated by the first periodic structure **140** does not propagate in the shortest straight distance from the $Z1$ side to the $Z2$ side of the photoelectric conversion unit **110**, but propagates in an oblique direction inside the photoelectric conversion unit **110**. Thus, the pixel **10** may increase the optical path length of light to be converted into an electric charge by the photoelectric conversion unit **110**, compared to a case in which the diffracted light Fr does not occur. In the pixel **10**, the amount of light absorbed in the photoelectric conversion unit **110** increases as the optical path length of light to be converted into an electric charge by the photoelectric conversion unit **110** increases. Due to this, the pixel **10** may exhibit high quantum efficiency in the photoelectric conversion unit **110**.

[0095] The period p of the first periodic structure **140** has a certain size based on the wavelength (the same meaning as the wavelength of incident light) and angle of incidence of light received by the photoelectric conversion unit **110**. In addition, the period p of the first periodic structure **140** may have a length that is shorter than the wavelength of light received, and appropriate for generating the diffracted light Fr in the photoelectric conversion unit **110**.

[0096] FIG. **6** illustrates an example of first-order diffracted light $Fr1$ and second-order diffracted light $Fr2$ both generated from light received by the first periodic structure **140**. As the period p is set to an appropriate length according to the angle of incidence and wavelength of the incident light F , the pixel **10** may generate the first-order diffracted light $Fr1$ and the second-order diffracted light $Fr2$ inside the first periodic structure **140**.

[0097] In the solid-state imaging device **1**, for example, in a case in which the angle of incidence of the incident light F is 0° and the wavelength of the incident light F is 940 nm, the length of the period p (in the cross-sectional view of FIG. **4A**, the straight distance between the centers of the second layers **142** adjacent to each other in the horizontal direction) is preferably 400 nm to 600 nm, and more preferably 600 nm. With the period p set in this manner, a diffraction angle $\theta1$ between the incident light $F1$ and the first-order diffracted light $Fr1$ is, for example, 25.7° , and a diffraction angle $\theta2$ between the incident light $F1$ and the second-order diffracted light $Fr2$ is, for example, 60.5° .

[0098] The length of the period p is not particularly limited as long as the photoelectric conversion unit **110** may generate the diffracted light Fr , considering the angle of incidence and wavelength of the incident light F . For example, even in a case in which the angle of incidence of the incident light F is not 0° or the wavelength of the incident light F is not 940 nm, the length of the period p may be changed to an appropriate length based on a known equation for diffraction of light (an equation for transmitted diffracted light). For example, in a case in which the angle of incidence of the incident light F is 30° and the wavelength is 940 nm, the length of the period p may be set within the range of 500 nm to 700 nm.

[0099] As illustrated in FIG. **4A**, the first periodic structure **140** may be configured such that the volume proportions of the first layers **141** decrease from the pixel central portion **11** toward the pixel peripheral portion **13**, because widths w of the second layers **142** in the horizontal direction are different from each other.

[0100] In the first periodic structure **140** illustrated in FIG. **4A**, the second layers **142a**, **142b**, **142c**, and **142d** have the same depth $d2$. In addition, in the first periodic structure **140**, the width w

gradually increases from the second layers **142a** arranged near the pixel central portion **11**, toward the second layers **142d** arranged in the pixel peripheral portion **13**. In the pixel **10**, the widths w of the respective second layers **142a**, **142b**, **142c**, and **142d** are different from each other as described above, and thus, the volume proportions of the first layers **141** may gradually decrease from the pixel central portion **11** toward the pixel peripheral portion **13**.

[0101] A depth $d1$ of the first layers **141** and the depth $d2$ of the second layers **142** in the stacking direction may be set to be equivalent to the wavelength of light received by the photoelectric conversion unit **110**. In addition, a thickness t of the photoelectric conversion unit **110** (the thickness in the stacking direction illustrated in FIG. 3) may be set to be greater than the depth $d1$ of the first layers **141** and the depth $d2$ of the second layers **142**.

[0102] In a case in which the wavelength of the incident light F is 940 nm as described above, the depth $d1$ of the first layers **141** and the depth $d2$ of the second layers **142** may be 940 nm, which is approximately the same as the wavelength of the incident light F . In addition, in a case in which the depth $d1$ of the first layers **141** and the depth $d2$ of the second layers **142** are 940 nm, the thickness t of the photoelectric conversion unit **110** may be, for example, 7000 nm.

[0103] In addition, the depth $d1$ of the first layers **141** and the depth $d2$ of the second layers **142** are not limited to 940 nm. For example, in a case in which the angle of incidence of the incident light F is 0° or 30° and the wavelength is 940 nm, the depth $d1$ of the first layers **141** and the depth $d2$ of the second layers **142** may be set to any value within the range of 100 nm to 940 nm.

[0104] As illustrated in FIGS. 3 and 7, the photoelectric conversion unit **110** is covered at least in part by a dielectric layer **146** having a lower refractive index than that of the first layers **141**. FIG. 7 is an enlarged view of a portion of the cross-sectional view of FIG. 3.

[0105] The dielectric layer **146** may be arranged to cover the photoelectric conversion unit **110** at three positions, for example, at left and right positions in the stacking direction (X1-side and X2-side positions), and at a Z2-side position. In some example embodiments, the dielectric layer **146** may contact external side and bottom surfaces of the first layers **141**. However, the position at which the dielectric layer **146** is arranged is not particularly limited, as long as an effect of totally reflecting diffracted light, which will be described below, may be exhibited.

[0106] For example, in a case in which the first layers **141** are formed of any one of the constituent materials described above, a constituent material of the dielectric layer **146** may be the same material as any one of the above-described examples of constituent materials of the second layers **142**. For example, the dielectric layer **146** may be formed of or include any one of silicon dioxide (SiO₂), silicon nitride (SiN), aluminum oxide (AlO), tantalum oxide (TaO), titanium nitride (TiN), and titanium oxide (TiO). However, the constituent material of the dielectric layer **146** is not particularly limited, as long as the dielectric layer **146** has a lower refractive index than that of the first layers **141**.

[0107] The period p of the first periodic structure **140** may be set to a length that allows the diffracted light Fr generated in the photoelectric conversion unit **110** to be totally reflected between the photoelectric conversion unit **110** and the dielectric layer **146**. FIG. 7 illustrates a state in which the diffracted light Fr is totally reflected by the dielectric layer **146**.

[0108] The pixel **10** may increase the optical path length of the diffracted light Fr generated inside the photoelectric conversion unit **110**, by totally reflecting the diffracted light Fr in the dielectric layer **146**. Thus, the pixel **10** may further increase the amount of light absorbed in the photoelectric conversion unit **110**.

[0109] As described above, in a case in which the incident light F has an angle of incidence of 0° and a wavelength of 940 nm, the first periodic structure **140** has a period p of 600 nm such that first-order diffracted light $Fr1$ having a diffraction angle $\theta1$ and second-order diffracted light $Fr2$ having a diffraction angle $\theta2$ may be generated in the photoelectric conversion unit **110** (see FIG. 6). In a case in which the period p has a length of 600 nm, a condition for the dielectric layer **146** to totally reflect each of the diffracted light $Fr1$ and the diffracted light $Fr2$ is that an angle of

incidence θ_3 (the angle of incidence θ_3 for the dielectric layer **146**) $\geq 23.7^\circ$ as illustrated in FIG. 7. This condition may be calculated by substituting the refractive index of the first layer **141** and the refractive index of the dielectric layer **146** into the known Snell's law.

[0110] The first periodic structure **140** may be configured such that the volume proportions of the first layers **141** and the second layers **142** are different from each other, based on the wavelength and angle of incidence of light received by the photoelectric conversion unit **110**. That is, the specific volume proportion of each layer **141** and **142** in each section in the horizontal direction may be arbitrarily set depending on the wavelength and angle of incidence of the incident light F. In a case in which the angle of incidence of the incident light F is 0° and the wavelength of the incident light F is 940 nm, the volume proportions of the first layers **141** near the pixel central portion **11** (the volume proportions of the first layers **141** with respect to the total volume of the photoelectric conversion unit **110**) may be, for example, 60% to 80%, and the volume proportions of the second layers **142** near the pixel central portion **11** (the volume proportions of the second layers **142** with respect to the total volume of the photoelectric conversion unit **110**) may be, for example, 20% to 40%. In a case in which the volume proportion of each layer **141** and **142** near the pixel central portion **11** is set as described above, the volume proportions of the first layers **141** near the pixel peripheral portion **13** may be, for example, 20% to 40%, and the volume proportions of the second layers **142** near the pixel peripheral portion **13** may be, for example, 60% to 80%. In addition, that the volume proportions of the first layers **141** and the second layers **142** may be set differently based on the wavelength and angle of incidence of the incident light F received by the photoelectric conversion unit **110** is also applicable to the volume proportions of third layers **153** and fourth layers **154** to be described below in a second embodiment.

[0111] A first wiring layer **131** arranged closest to the photoelectric conversion unit **110** in the stacking direction has a reflective structure that reflects light transmitted through the photoelectric conversion unit **110**.

[0112] In the pixel **10**, the first wiring layer **131** has the reflective structure, and thus, at least part of transmitted light F_t , which has passed through the photoelectric conversion unit **110** and reached the first wiring layer **131**, may be reflected toward the photoelectric conversion unit **110**. In the pixel **10**, light reflected from the reflective structure of the first wiring layer **131** enters the photoelectric conversion unit **110**, and thus, the optical path length of light inside the photoelectric conversion unit **110** may be further increased. Accordingly, the pixel **10** may further increase the amount of light absorbed in the photoelectric conversion unit **110**.

[0113] The reflective structure installed in the first wiring layer **131** may be configured by, for example, a metal layer installed on at least part of the surface of the first wiring layer **131** arranged on the side of the photoelectric conversion unit **110**. The metal layer may be formed of, for example, tungsten, aluminum, copper, etc.

[0114] Hereinafter, modified examples regarding the horizontal arrangement of each layer **141** and **142**, or the cross-sectional shape of each layer **141** and **142** in the stacking direction will be described with reference to FIGS. 8 to 15. FIGS. 8 to 10 are cross-sectional views of respective modified examples corresponding to the cross-sectional view of FIG. 4, and FIGS. 11 to 15 are plan views of respective modified examples corresponding to the plan view of FIG. 2.

[0115] As will be described about each of the following modified examples, the cross-sectional shape of each layer **141** and **142** of the first periodic structure **140** in the stacking direction and the arrangement of each layer **141** and **142** in plan view may be arbitrarily changed as long as the lens effect and the effect of generating diffracted light described above may be achieved.

[0116] As illustrated in FIG. 8, the first periodic structure **140** of modified example 1 is configured such that the volume proportions of the first layers **141** decrease from the pixel central portion **11** toward the pixel peripheral portion **13**, because the depths d_2 of the second layers **142** in the stacking direction are different from each other.

[0117] In the first periodic structure **140** illustrated in FIG. 8, the widths w of the respective second

layers **142a**, **142b**, **142c**, and **142d**) are equal to each other. In addition, in the first periodic structure **140**, the depths **d2** gradually increase from the second layers **142a** arranged near the pixel central portion **11**, to the second layers **142d** arranged near the pixel peripheral portion **13**. In the pixel **10**, the depths **d2** of the respective second layers **142a**, **142b**, **142c**, and **142d** are different from each other, and thus, the volume proportions of the first layers **141** may gradually decrease from the pixel central portion **11** toward the pixel peripheral portion **13**.

[0118] As illustrated in FIG. **9**, the first periodic structure **140** of modified example 2 is configured such that the volume proportions of the first layers **141** decrease from the pixel central portion **11** toward the pixel peripheral portion **13**, because the second layers **142** differ from each other in the width **w** in the horizontal direction and in the depth **d2** in the stacking direction.

[0119] In the first periodic structure **140** illustrated in FIG. **9**, the widths **w** and the depths **d2** gradually increase from the second layers **142a** arranged near the pixel central portion **11**, to the second layers **142d** arranged near the pixel peripheral portion **13**. In the pixel **10**, the second layers **142a**, **142b**, **142c**, and **142d** differ from each other in the width **w** and the depth **d2**, and thus, the volume proportions of the first layers **141** may gradually decrease from the pixel central portion **11** toward the pixel peripheral portion **13**.

[0120] As illustrated in FIG. **10**, in the first periodic structure **140** of modified example 3, the second layer **142** has a triangular cross-sectional shape with a base arranged on the **Z1** side and a vertex arranged on the **Z2** side.

[0121] As illustrated in the present modified example, the second layer **142** may be configured to have any cross-sectional shape as long as the magnitude relationship between the volume proportions of the respective layers **141** and **142** in the horizontal direction may be adjusted.

[0122] In the first periodic structure **140** of modified example 3, the volume proportions of the first layers **141** decrease from the pixel central portion **11** toward the pixel peripheral portion **13**, because the second layers **142** differ from each other in the width **w** in the horizontal direction and the depth **d2** in the stacking direction. In addition, in a case in which the cross-sectional shape of the second layer **142** is the shape of a triangle as illustrated in modified example 3, the width **w** may be defined as the length of the base of the triangle.

[0123] As illustrated in FIG. **11**, in the pixel **10** of modified example 4, the first layers **141** and the second layers **142** are arranged in the shape of a square frame in plan view. For example, each of the first layers **141** and the second layers **142** may extend continuously to concentrically surround the pixel central portion **11**. Each of the first layers **141** may become narrower as the distance increases from the pixel central portion **11**, and each of the second layer **142** may become wider as the distance increases from the pixel central portion **11**. When viewed in cross-section, in the pixel **10** of modified example 4, the first layers **141** and the second layers **142** may include features of any one of modified examples 1 to 3 of FIGS. **8** to **10**, respectively.

[0124] As illustrated in FIG. **12**, in the pixel **10** of modified example 5, the second layers **142** are arranged in the shapes of rectangles with certain lengths and widths in plan view. In addition, in the first periodic structure **140** of modified example 5, the first layers **141** are arranged at portions corresponding to vertices of a square formed by the second layers **142** in plan view. When viewed in cross-section, in the pixel **10** of modified example 5, the first layers **141** and the second layers **142** may include features of any one of modified examples 1 to 3 of FIGS. **8** to **10**, respectively.

[0125] As illustrated in FIG. **13**, in the pixel **10** of modified example 6, the second layers **142** are arranged in the shapes of rectangles with certain lengths and widths in plan view. The lengths of the second layers **142** in each direction of arrows **A1** and **A2** are less than the lengths of the second layers **142** of modified example 5. That is, the second layers **142** of modified example 6 has a structure in which the second layers **142** of modified example 5 are subdivided in plan view. When viewed in cross-section, in the pixel **10** of modified example 6, the first layers **141** and the second layers **142** may include features of any one of modified examples 1 to 3 of FIGS. **8** to **10**, respectively.

[0126] As illustrated in FIG. 14, in the pixel 10 of modified example 7, the second layers 142 are arranged in the shapes of rectangles with certain lengths and widths in plan view. The second layers 142 of modified example 7 are formed such that some of the second layers 142 of modified example 6 are connected to each other in plan view. When viewed in cross-section, in the pixel 10 of modified example 7, the first layers 141 and the second layers 142 may include features of any one of modified examples 1 to 3 of FIGS. 8 to 10, respectively.

[0127] As illustrated in FIG. 15, in the pixel 10 of modified example 8, the second layers 142 are arranged to be connected to each other in the longitudinal and transverse directions in plan view, like the squares of a checkerboard. When viewed in cross-section, in the pixel 10 of modified example 8, the first layers 141 and the second layers 142 may include features of any one of modified examples 1 to 3 of FIGS. 8 to 10, respectively.

[0128] In addition, the first periodic structure 140 may be arbitrarily changed as long as it has a certain period p across the entire area or a partial area in the horizontal direction, in addition to the arrangement of each layer 141 and 142 described above regarding modified examples 4 to 8. For example, each layer 141 and 142 may be arranged in a concentric pattern with the center arranged in the pixel central portion 11 in plan view.

[0129] As described above, the solid-state imaging device 1 according to the first embodiment is a solid-state imaging device including the pixel array 20 consisting of a plurality of pixels 10, wherein the pixel 10 includes the photoelectric conversion unit 110 configured to convert light into an electric charge, the on-chip lens 120 installed on the Z1 side of the photoelectric conversion unit 110, the first wiring layer 131 installed on the Z2 side of the photoelectric conversion unit 110, and the first periodic structure 140 that is installed on the Z1 side of the photoelectric conversion unit 110 and has periodicity in a horizontal direction that is perpendicular to a stacking direction of the photoelectric conversion unit 110, and the first periodic structure 140 includes a plurality of first layers 141 and a plurality of second layers 142 having a lower refractive index than that of the first layers 141. In at least one of the plurality of pixels 10, the first periodic structure 140 may be configured such that the volume proportions of the first layers 141 in the pixel central portion 11 in the horizontal direction are greater than the volume proportions of the second layers 142 in the pixel central portion 11, and the first layers 141 and the second layers 142 may be arranged such that the volume proportions of the first layers 141 decrease from the pixel central portion 11 toward the pixel peripheral portion 13 in the horizontal direction.

[0130] In the pixel 10 of the solid-state imaging device 1 of the present embodiment, the first periodic structure 140 installed on the Z1 side of the photoelectric conversion unit 110 may exhibit a lens effect that collects the incident light F entering through the on-chip lens 120, onto the photoelectric conversion unit 110, and may generate a larger amount of diffracted light F_r inside the photoelectric conversion unit 110. Thus, the solid-state imaging device 1 may reduce color mixing and improve light absorption in the photoelectric conversion unit 110.

[0131] Hereinafter, solid-state imaging devices according to second to fifth embodiments will be described. In the description of the second to fifth embodiments, descriptions of members, configurations, or the like provided above in the first embodiment will not be repeated. In addition, those not specifically described in the second to fifth embodiments may be the same as those in the first embodiment.

Second Embodiment

[0132] FIG. 16 is a cross-sectional view of pixels 10A according to a second example embodiment, taken along a stacking direction. FIG. 17 is a cross-sectional view for describing functional effects of the pixel 10A.

[0133] As illustrated in FIG. 16, the pixel 10A according to the second embodiment includes the first periodic structure 140 and a second periodic structure 150 that is installed on the Z2 side of the photoelectric conversion unit 110 and has periodicity in a horizontal direction perpendicular to the stacking direction.

[0134] The second periodic structure **150** includes the third layers **153** having the same refractive index as that of the first layers **141**, and the fourth layers **154** having the same refractive index as that of the second layers **142**.

[0135] The third layer **153** may be formed of, for example, the same material as the material exemplified above as the constituent material of the first layer **141**. The fourth layer **154** may be formed of, for example, the same material as the material exemplified above as the constituent material of the second layer **142**. In a case in which the third layers **153** and the fourth layers **154** are configured as described above, the first layers **141** and the third layer **153** have the same optical properties (e.g., the refractive index), and the second layers **142** and the fourth layers **154** also have the same optical properties (e.g., the refractive index). Thus, the difference in the refractive index between the third layers **153** and the fourth layers **154** is the same as the difference in the refractive index between the first layers **141** and the second layers **142**.

[0136] The functional effects (improvement of quantum efficiency by reflection of diffracted light) of the second periodic structure **150** will be described with reference to FIG. **17**. In addition, FIG. **17** is a schematic diagram for describing a principle of reflection of diffracted light, and the arrangement or cross-sectional shapes of the third layers **153** and the fourth layers **154** illustrated in FIG. **17** do not have a strict correspondence with the embodiment illustrated in FIG. **16**.

[0137] In the pixel **10A**, when light that has passed through the photoelectric conversion unit **110** reaches the second periodic structure **150** arranged on the Z2 side of the photoelectric conversion unit **110**, diffracted light Fr is generated at a boundary between the photoelectric conversion unit **110** and the second periodic structure **150**. In addition, FIG. **17** illustrates an example of first-order diffracted lights Fr1' generated from incident lights F2 and F3 collected by the on-chip lens **120**, and second-order diffracted lights Fr2' generated from diffracted lights Fr1 and Fr2 reflected from the dielectric layer **146** (see FIG. 7).

[0138] In the pixel **10A**, as the diffracted light Fr is generated in the second periodic structure **150**, light that has reached the second periodic structure **150** is moved from the Z2 side of the photoelectric conversion unit **110** toward the Z1 side. Thus, the pixel **10A** may further increase the optical path length of light to be converted into an electric charge by the photoelectric conversion unit **110**. Therefore, the pixel **10A** may further increase the amount of light absorbed in the photoelectric conversion unit **110**.

[0139] As illustrated in FIG. **16**, the second periodic structure **150** may be configured to have the same periodicity as that of the first periodic structure **140**. That is, the length of the period p of the first periodic structure **140** and the length of the period p of the second periodic structure **150** may be set to be approximately equal to each other. As described above, the length of the period p of the first periodic structure **140** may be set to 600 nm, based on the angle of incidence (e.g., 0°) and wavelength (e.g., 940 nm) of the incident light F. Thus, in the present embodiment, the length of the period p of the second periodic structure **150** is set to 600 nm.

[0140] In the pixel **10A**, the length of the period p of the first periodic structure **140** and the length of the period p of the second periodic structure **150** are set to be approximately equal to each other, and thus, diffraction angles for diffracted lights Fr generated on the Z1 side and the Z2 side of the photoelectric conversion unit **110**, respectively, may be approximately equal to each other. In the pixel **10A**, the diffraction angle of the diffracted lights Fr generated on the Z1 side and the Z2 side of the photoelectric conversion unit **110** are fixed to be approximately equal to each other, making it easy to design an optical path for total reflection of light inside the photoelectric conversion unit **110**. In addition, in the pixel **10A**, the length of the period p of the second periodic structure **150** arranged on the Z2 side of the photoelectric conversion unit **110** is set to 600 nm, which is equal to the length of the period p of the first periodic structure **140**, and thus, light that has reached the second periodic structure **150** may be totally reflected.

[0141] In the example illustrated in FIG. **17**, a diffraction angle θ_4 of the first-order diffracted light Fr1' generated in the second periodic structure **150** is approximately 60°, which is equal to the

diffraction angle $\theta 1$ of the first-order diffracted light Fr1 generated in the first periodic structure **140** (see FIG. 6). In addition, a diffraction angle $\theta 5$ of the second-order diffracted light Fr2' generated in the second periodic structure **150** is approximately 26° , which is equal to the diffraction angle $\theta 2$ of the second-order diffracted light Fr2 generated in the first periodic structure **140** (see FIG. 6).

[0142] In the pixel **10A**, the distribution of the volume proportions of the first layers **141** in the first periodic structure **140** is the same as the distribution of the volume proportions of the third layers **153** in the second periodic structure **150**. In addition, the distribution of the volume proportions of the second layers **142** in the first periodic structure **140** is the same as the distribution of the volume proportions of the fourth layers **154** in the second periodic structure **150**.

[0143] In the pixel **10A**, as described above, because the distribution of the volume proportions of the first layers **141** is the same as the distribution of the volume proportions of the third layers **153**, and the distribution of the volume proportions of the second layers **142** is the same as the distribution of the volume proportions of the fourth layers **154**, the diffraction angles of the diffracted lights Fr generated on the Z1 side and the Z2 side of the photoelectric conversion unit **110** are further fixed. Thus, in the pixel **10A**, the design of an optical path for totally reflecting light inside the photoelectric conversion unit **110** is further facilitated.

[0144] FIG. 16 shows the volume proportion of the first periodic structure **140** in the horizontal direction, and the cross-sectional shape of each layer **141** and **142**, as an example of the configuration illustrated in FIGS. 4A and 4B. In addition, in FIG. 16, the second periodic structure **150** is configured to have the same volume proportion and cross-sectional shape as those of the first periodic structure **140**. However, when the pixel **10A** is configured to have the first periodic structure **140** and the second periodic structure **150**, there are no particular limitations on the horizontal volume proportion or cross-sectional shape of each of the first and second periodic structures **140** and **150**. For example, each of the first and second periodic structures **140** and **150** may adopt any one of the configurations described above in modified examples 1 to 8.

Third Embodiment

[0145] FIG. 18 is a cross-sectional view of pixels **10B** according to a third example embodiment, taken along a stacking direction. FIG. 19 includes a graph showing the relationship between the magnitude of an average refractive index for light F incident on the photoelectric conversion unit **110**, and horizontal positions in the photoelectric conversion unit **110**, and a schematic cross-sectional view of the first periodic structure **140** taken along a stacking direction.

[0146] As illustrated in FIGS. 18 and 19, in the pixel **10B** according to the third embodiment, a position at which the first layer **141** of the first periodic structure **140** has the largest volume proportion, is arranged at a position corresponding to the position of the central portion **121** (a central portion in the horizontal direction) of the on-chip lens **120**.

[0147] The central portion **121** of the on-chip lens **120** may be arranged at a certain position between the pixel central portion **11** and the pixel peripheral portion **13**, in correspondence with the angle of incidence of light received by the photoelectric conversion unit **110**.

[0148] The examples illustrated in FIGS. 18 and 19 show a state in which light is incident at a certain angle of incidence from the left side in the drawings (the left side in the horizontal direction).

[0149] The central portion **121** of the on-chip lens **120** is arranged at a position displaced toward the side from which the light is coming. When light is incident on the on-chip lens **120** at a certain angle of incidence, in order to efficiently collect the incident light F3 onto the first wiring layer **131** arranged on the Z2 side of the photoelectric conversion unit **110**, it is preferable that the incident light F3, which is coming from a more rightward position in the horizontal direction with respect to the central portion **121** than the incident light F1, be further refracted toward the center by the first periodic structure **140**. Thus, in the pixel **10B**, the position of the central portion **121** of the on-chip lens **120** is set to a position shifted to the left in the horizontal direction, and the position where the

volume proportion of the first layer **141** of the first periodic structure **140** is the largest (the positions indicated by imaginary lines H1 in FIG. **19**) is arranged to correspond to the position of the central portion **121** of the on-chip lens **120**.

[0150] In addition, the two-dot chain line in the graph of FIG. **19** represents the distribution of refractive indices in the horizontal direction in a case in which the angle of incidence of the incident light F is 0° (the distribution of refractive indices of the first embodiment shown in FIG. **5**). The imaginary line H1 in the graph of FIG. **19** indicates a position in the first periodic structure **140** where the refractive index is the largest (the same position as the pixel central portion **11**), as described above in the first embodiment.

[0151] In the pixel **10B** of the third embodiment, a position at which the volume proportion of the first layer **141** of the first periodic structure **140** is the largest, is arranged to correspond to the position of the central portion **121** of the on-chip lens **120**. Thus, even when light F is incident on the first periodic structure **140** at a certain angle of incidence, the light may be efficiently collected onto the first wiring layer **131** arranged on the Z2 side of the photoelectric conversion unit **110**.

Fourth Embodiment

[0152] FIG. **20** is a cross-sectional view of pixels **10C** according to a fourth example embodiment, taken along a stacking direction. FIG. **21** includes a graph showing the relationship between the magnitude of an average refractive index for light incident on the photoelectric conversion unit **110**, and horizontal positions in the photoelectric conversion unit **110**, and a schematic cross-sectional view of the first periodic structure **140** taken along a stacking direction.

[0153] In at least one of a plurality of pixels **10** included in the solid-state imaging device **1**, the first periodic structure **140** may adopt a configuration in which, as illustrated in FIGS. **20** and **21**, the first layers **141** and the second layers **142** are arranged such that, in intermediate regions **17** between the pixel central portion **11** and the pixel peripheral portion **13**, the volume proportions of the first layers **141** are greater than the volume proportions of the second layers **142**, and the volume proportions of the first layers **141** decrease from the intermediate region **17** toward the pixel central portion **11** and the pixel peripheral portion **13**.

[0154] In the example illustrated in FIG. **21**, one intermediate region **17** is set between the pixel central portion **11** in the horizontal direction and the pixel peripheral portion **13** that is located on one side (left) in the horizontal direction. In addition, in this example, one intermediate region **17** is set between the pixel central portion **11** in the horizontal direction and the pixel peripheral portion **13** located on the other side (right) in the horizontal direction.

[0155] As described above, because, in the first periodic structure **140**, the volume proportions of the first layers **141** in the intermediate region **17** are greater than the volume proportions of the second layers **142** in the intermediate region **17**, and the volume proportions of the first layers **141** decrease from the intermediate region **17** toward the pixel central portion **11** and the pixel peripheral portion **13**, the average refractive index in the horizontal direction is the largest at positions corresponding to two intermediate regions **17**, as shown in the graph of FIG. **21**.

[0156] In the pixel **10C** including the first periodic structure **140** configured as described above, even in a case in which, for example, the first wiring layers **131** arranged on the Z2 side of the photoelectric conversion unit **110** are arranged not to overlap with the central portion **121** of the on-chip lens **120** as illustrated in FIG. **20** (in a case in which the first wiring layer **131** is arranged at a position shifted in the horizontal direction from the central portion **121** of the on-chip lens **120**), the incident light F1 passing through the pixel central portion **11** (the central portion **121** of the first periodic structure **140**), and other incident lights F2 and F3 may be collected from their positions in the two intermediate regions **17**, onto each of the first wiring layers **131** arranged at certain intervals in the horizontal direction. Thus, the pixel **10C** may effectively prevent the occurrence of color mixing even in a case in which it has a stack structure in which the first wiring layers **131** are arranged not to overlap with the central portion **121** of the on-chip lens **120**.

[0157] In addition, in the present embodiment, when installing two intermediate regions **17** where

the refractive index of one first periodic structure **140** is the largest, as illustrated in FIG. **21**, the widths w of the second layers **142** may be set to increase from the intermediate regions **17** toward the pixel central portion **11**, and may also be set to increase from the intermediate regions **17** toward the pixel peripheral portion **13**.

Fifth Embodiment

[0158] FIG. **22** is a cross-sectional view of pixels **10D** according to a fifth example embodiment, taken along a stacking direction. FIG. **23** includes a graph showing the relationship between the magnitude of an average refractive index for light incident on the photoelectric conversion unit **110**, and horizontal positions in the photoelectric conversion unit **110**, and a schematic cross-sectional view of the first periodic structure **140** taken along a stacking direction.

[0159] In at least one of a plurality of pixels **10D** included in the solid-state imaging device **1**, the first periodic structure **140** may adopt a configuration in which, as illustrated in FIGS. **22** and **23**, the first layers **141** and the second layers **142** are arranged such that, in proximity regions **15** close to the pixel peripheral portion **13**, the volume proportions of the first layers **141** is greater than the volume proportions of the second layers **142**, and the volume proportions of the first layers **141** decrease from the pixel peripheral portion **13** toward the pixel central portion **11**.

[0160] The first periodic structure **140** may be configured such that, in the proximity regions **15** close to the pixel peripheral portion **13**, the volume proportions of the first layers **141** are greater than the volume proportions of the second layers **142**, and the volume proportions of the first layers **141** decrease from the pixel peripheral portion **13** toward the pixel central portion **11**. Thus, in the first periodic structure **140**, the average refractive index may be the smallest at a position corresponding to the pixel central portion **11**, as shown in the graph of FIG. **23**. In addition, an imaginary line H3 in the graph of FIG. **19** indicates a position in the first periodic structure **140** where the refractive index is the smallest (the same position as the pixel central portion **11**).

[0161] In the pixel **10D** including the first periodic structure **140** configured as described above, even in a case in which, for example, the first wiring layers **131** arranged on the Z2 side of the photoelectric conversion unit **110** are arranged not to overlap with the central portion **121** of the on-chip lens **120** as illustrated in FIG. **22**, the incident light F1 passing through the pixel central portion **11** (the central portion **121** of the first periodic structure **140**), and other incident lights F2 and F3 may be collected from their positions in the two intermediate regions **17**, onto each of the first wiring layers **131** arranged at certain intervals in the horizontal direction. Thus, the pixel **10D** may effectively prevent the occurrence of color mixing even in a case in which it has a stack structure in which the first wiring layers **131** are arranged not to overlap with the central portion **121** of the on-chip lens **120**.

[0162] In addition, in the present embodiment, when the refractive index of a region corresponding to the pixel central portion **11** is the smallest and the refractive index of two regions corresponding to the pixel peripheral portion **13** is the largest, the widths w of the second layers **142** are set to decrease from the pixel central portion **11** toward the pixel peripheral portion **13**, as illustrated in FIG. **23**. The period p may be set to have a certain length in the horizontal direction of the first periodic structure **140**.

[0163] The solid-state imaging device according to the inventive concept has been described through a plurality of embodiments and a plurality of modified examples, but the inventive concept is not limited to the descriptions provided herein and may be appropriately modified.

[0164] For example, in the third to fifth embodiments, examples of pixels each including a first periodic structure and a second periodic structure are provided, but the pixels may be configured to include only the first periodic structure. In addition, the cross-sectional shapes in the stacking direction or the arrangement in plan view of each layer of the first periodic structure and/or each layer of the second periodic structure according to the first to fifth embodiments may be configured by arbitrarily adopting and combining those of modified examples 1 to 8.

[0165] While the inventive concept has been particularly shown and described with reference to

embodiments thereof, it will be understood that various changes in form and details may be made therein without departing from the spirit and scope of the following claims.

Claims

1. A solid-state imaging device comprising a pixel array comprising a plurality of pixels, wherein each of the plurality of pixels comprises: a photoelectric conversion unit configured to convert light into an electric charge; an on-chip lens installed on one side of the photoelectric conversion unit; wiring layers installed on another side of the photoelectric conversion unit; and a first periodic structure that is installed on the one side of the photoelectric conversion unit and has periodicity in a horizontal direction that is perpendicular to a stacking direction of the photoelectric conversion unit, wherein the first periodic structure comprises a plurality of first layers and a plurality of second layers that have refractive indices lower than refractive indices of the plurality of first layers, and wherein in at least one of the plurality of pixels, the first layers and the second layers of the first periodic structure are arranged such that volume proportions of the first layers in a pixel central portion are greater than volume proportions of the second layers in the pixel central portion, and the volume proportions of the first layers decrease from the pixel central portion toward a pixel peripheral portion in the horizontal direction.
2. The solid-state imaging device of claim 1, wherein widths of the second layers in the horizontal direction are different from each other, and wherein the first periodic structure is configured such that the volume proportions of the first layers decrease from the pixel central portion toward the pixel peripheral portion.
3. The solid-state imaging device of claim 1, wherein depths of the second layers in the stacking direction are different from each other, and wherein the first periodic structure is configured such that the volume proportions of the first layers decrease from the pixel central portion toward the pixel peripheral portion.
4. The solid-state imaging device of claim 1, wherein a period of the first periodic structure is set to a certain value based on a wavelength and an angle of incidence of light received by the photoelectric conversion unit, and comprises a length that is less than the wavelength of the received light, and enables generation of diffracted light in the photoelectric conversion unit.
5. The solid-state imaging device of claim 1, wherein depths of the first layers and depths of the second layers in the stacking direction are equal to a wavelength of light received by the photoelectric conversion unit, and wherein a thickness of the photoelectric conversion unit is greater than the depths of the first layers and the depths of the second layers.
6. The solid-state imaging device of claim 1, wherein the photoelectric conversion unit is covered at least in part by a dielectric layer having a refractive index lower than the refractive indices of the first layers, and wherein a period of the first periodic structure comprises a length that enables diffracted light, which is generated in the photoelectric conversion unit, to be totally reflected at a boundary between the photoelectric conversion unit and the dielectric layer.
7. The solid-state imaging device of claim 1, wherein a wiring layer, which is arranged at a position closest to the photoelectric conversion unit in the stacking direction, of the wiring layers has a reflective structure that reflects light transmitted through the photoelectric conversion unit.
8. The solid-state imaging device of claim 1, wherein a position where a volume proportion of a first layer of the first layers of the first periodic structure is largest is arranged at a position corresponding to a position of a central portion of the on-chip lens in the horizontal direction, and wherein the central portion of the on-chip lens is arranged at a certain position between the pixel central portion and the pixel peripheral portion, in correspondence with an angle of incidence of light received by the photoelectric conversion unit.
9. The solid-state imaging device of claim 1, wherein, in at least one of the plurality of pixels, the first layers and the second layers of the first periodic structure are arranged such that, in an

intermediate region between the pixel central portion and the pixel peripheral portion, the volume proportions of the first layers are greater than the volume proportions of the second layers, and the volume proportions of the first layers decrease from the intermediate region toward the pixel central portion and the pixel peripheral portion.

10. The solid-state imaging device of claim 1, wherein, in at least one of the plurality of pixels, the first layers and the second layers of the first periodic structure are arranged such that, in a proximity region close to the pixel peripheral portion, the volume proportions of the first layers are greater than the volume proportions of the second layers, and the volume proportions of the first layers decrease from the pixel peripheral portion toward the pixel central portion.

11. The solid-state imaging device of claim 1, wherein the photoelectric conversion unit comprises a plurality of photoelectric conversion units, and wherein the plurality of the photoelectric conversion units are separated from each other by an insulating film.

12. The solid-state imaging device of claim 1, wherein the photoelectric conversion unit comprises a plurality of photoelectric conversion units, wherein the plurality of the photoelectric conversion units are separated from each other by a fixed charge film, and wherein the fixed charge film comprises an oxide film or a nitride film comprising at least one metal element from among hafnium (Hf), aluminum (Al), zirconium (Zr), tantalum (Ta), and titanium (Ti).

13. A solid-state imaging device comprising a pixel array comprising a plurality of pixels, wherein each of the plurality of pixels comprises: a photoelectric conversion unit configured to convert light into an electric charge; an on-chip lens installed on one side of the photoelectric conversion unit; a wiring layer installed on another side of the photoelectric conversion unit; and a first periodic structure that is installed on the one side of the photoelectric conversion unit and has periodicity in a horizontal direction that is perpendicular to a stacking direction of the photoelectric conversion unit, wherein the first periodic structure comprises a plurality of first layers and a plurality of second layers that have refractive indices lower than refractive indices of the plurality of first layers, wherein in at least one of the plurality of pixels, the first layers and the second layers of the first periodic structure are arranged such that volume proportions of the first layers in a pixel central portion are greater than volume proportions of the second layers in the pixel central portion, and the volume proportions of the first layers decrease from the pixel central portion toward a pixel peripheral portion in the horizontal direction, wherein the pixel comprising the first periodic structure further comprises a second periodic structure that is installed on the one side of the photoelectric conversion unit and has periodicity in the horizontal direction, and wherein the second periodic structure comprises a plurality of third layers having refractive indices equal to the refractive indices of the first layers, and a plurality of fourth layers having refractive indices equal to the refractive indices of the second layers.

14. The solid-state imaging device of claim 13, wherein the second periodic structure has periodicity that is identical to the periodicity of the first periodic structure.

15. The solid-state imaging device of claim 13, wherein distribution of the volume proportions of the first layers in the first periodic structure is identical to distribution of volume proportions of the third layers in the second periodic structure, and wherein distribution of the volume proportions of the second layers in the first periodic structure is identical to distribution of volume proportions of the fourth layers in the second periodic structure.

16. The solid-state imaging device of claim 13, wherein at least one of the first periodic structure and the second periodic structure is configured based on a wavelength and an angle of incidence of light received by the photoelectric conversion unit, such that at least one of the volume proportions of the first layers and the second layers and volume proportions of the third layers and the fourth layers are different from each other.

17. The solid-state imaging device of claim 13, wherein at least one of a period of the first periodic structure and a period of the second periodic structure in a pixel array central portion differs from that in a pixel array peripheral portion.

18. A solid-state imaging device comprising a pixel array comprising a plurality of pixels, wherein each of the plurality of pixels comprises: a photoelectric conversion unit configured to convert light into an electric charge; an on-chip lens installed on one side of the photoelectric conversion unit; a wiring layer installed on another side of the photoelectric conversion unit; and a first periodic structure that is installed on the one side of the photoelectric conversion unit and has periodicity in a horizontal direction that is perpendicular to a stacking direction of the photoelectric conversion unit, wherein the first periodic structure comprises a plurality of first layers and a plurality of second layers that have refractive indices lower than refractive indices of the plurality of first layers, and wherein in at least one of the plurality of pixels, depths of the second layers of the first periodic structure are equal to each other, and widths of the second layers increase from a pixel central portion toward a pixel peripheral portion.

19. The solid-state imaging device of claim 18, wherein the first layers comprise any one of silicon (Si), germanium (Ge), and indium gallium arsenide (InGaAs).

20. The solid-state imaging device of claim 18, wherein the second layers comprise any one of silicon dioxide (SiO₂), silicon nitride (SiN), aluminum oxide (AlO), tantalum oxide (TaO), titanium nitride (TiN), and titanium oxide (TiO).
